Operation and Maintenance Manual

Electric Drilling Package 50Hz with
MTU Series 4000 12 V 4000 G23
TPDSZ12V4000-2A0
Oil & Gas

SS180038/00E

Power. Passion. Partnership.

Preface

This Operation & Maintenance Manual provides general instructions for operating your MTU engine-generator set properly. It is essential that every person who works on or with the engine-generator set be completely familiar with the contents of this manual and that he/she carefully follows the instructions contained herein.

Each operation and maintenance may require some modification of the suggested guidelines in this manual. They must be consistent with locally applicable standards and take into consideration safety guidelines and measures.

Following this guide will result in an efficient and reliable installation.

Carefully follow all procedures and safety precautions to ensure proper equipment operation and to avoid bodily injury. Read and follow the Safety Precautions section at the beginning of this manual.

All instructions and diagrams have been checked for accuracy and simplicity of application, however, the skills of the operator are most important. MTU does not guarantee the result of any operation contained in this manual nor can MTU assume responsibility for any injury or damage to property. Persons engaging in operating do so entirely at their own risk.

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1.1 General Service Safety

Contact Authorized Service for Additional Help

If you have any questions about a procedure or safety message related to servicing this engine, contact an authorized service outlet for assistance. Repair or engine overhaul must be carried out in an MTU authorized workshop.

Trained and Authorized Personnel only

Maintenance and repair work is to be carried out by trained and authorized personnel only. Do not allow unauthorized personnel on or near the engine when the engine is serviced. Work on or around this engine must be performed by people who have the necessary training, skills and tools to do the work. Improper maintenance or repair work can cause severe injury or death for yourself or bystanders.

Using this Manual

Read and understand this safety section before working on this product. This section contains general safety messages. Additional safety messages are included in the specific task procedures.

Signal Words Used in this Manual

Failure to follow the signal words and safety alert symbols contained in this manual can result in serious injury or death. Important safety information is distinguished in this manual by the following notations:

DANGER	In the event of immediate danger. **Consequences: Death or serious injury** • Remedial action
WARNING	In the event of potentially dangerous situations. **Consequences: Death or serious injury** • Remedial action
CAUTION	In the event of hazardous situations. **Consequences: Minor or moderate injuries** • Remedial action
NOTICE	In the event of a situation involving potentially adverse effects on the product. **Consequences: Material damage!** • Remedial action. • Additional product information.

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Consider the Safety of Procedures You Choose

The manufacturer cannot anticipate every possible circumstance that might involve a potential hazard. If a tool, procedure, work method or operating technique that is not specifically recommended by the manufacturer is used, you must consider possible hazards and satisfy yourself that it is safe for you and for others. Ensure the product will not be damaged or made unsafe by the transport, installation, commissioning, operation, lubrication, maintenance, repair or storage procedures that you choose.

Use Genuine Parts and do not make Unauthorized Modifications

Unauthorized modifications, third-party add-on devices and replacement parts may compromise the integrity of the equipment, and may present a safety risk. Only genuine MTU parts are allowed to be used to replace components or assemblies. Failure to heed this warning can lead to product damage, serious injury or death and may cause non-compliance with environmental regulations.

Follow Applicable Standards and Regulations

MTU recommends owners, operators and certified technicians become familiar with applicable national safety standards. To reduce the risk of severe injury or death, follow all codes, standards, regulations and laws pertaining to the products installation and application.

Maintain Product Labels

Safety labels should be legible and not covered. If any safety labels are damaged, missing or illegible, contact an MTU-authorized distributor or dealer to obtain replacement labels.

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1.2 Use of MTU Technical Documentation

Access to MTU technical documentation is restricted, due to the highly technical nature of the information. Maintenance and workshop documentation is structured based on Qualification Levels (QLs). Personnel are to be qualified at the appropriate level to carry out specific tasks. The QLs are matched to MTU training courses and a demonstrated proficiency related to various tool kits.

QL1 - Operational monitoring and maintenance tasks that can be done during breaks in operation, and do not require disassembly of the system, engine or any components.

QL2 - Maintenance procedures requiring the exchange of assemblies.

QL3 - Maintenance procedures requiring overhaul of assemblies.

Do not use MTU Technical Documentation or attempt to perform the activities described in this manual if you have not been properly qualified by an MTU-authorized distributor for this QL level.

The information, specifications and illustrations in this manual are based upon information available at the time it was written. The information, specifications and illustrations can change at any time. These changes can affect the service given to the product. Obtain the complete and most current information before you start any work on the system, engine or any components.

If you did not receive this manual through MTU, stop all work and contact MTU to make sure you have the appropriate training and qualifications to follow the procedures within.

Keep this Manual near Engine for Personnel [QL1]

This publication must be provided to all MTU-certified personnel involved in any work being done on the engine and a copy must be located in the vicinity of the engine, accessible at all times.

General Responsibilities of Owners and Operators and MTU Technicians [QL1]

<u>Owner and operator responsibilities</u>

In preparation for service of an engine, owners and operators must:

- Keep unauthorized personnel clear of engine service areas.
- Provide a clear path for transport of the engine.
- Provide a service area for the engine.
- Provide proper equipment to transport the engine.
- Provide that lifting and handling of heavy objects are performed or supervised by personal knowledgeable with lifting procedures.
- Provide certified forklift and crane operators.
- Provide proper tools for working on an engine that are properly maintained and calibrated.
- Provide necessary add-on heat generating devices, such as a fuel heater.
- Make sure that servicing of batteries is performed or supervised by personnel knowledgeable with batteries and required battery service procedures.

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1.3 MTU Engine Safety Section

Preparing to Service Engine

Shut off equipment

You could be seriously injured or killed when working on running equipment. A release or spill of coolant, oil, fuel or hydraulic fluid while the engine is running can cause a fire or explosion. Do not fill coolant or fuel tanks while the engine is running. To reduce the risk of exposure to compressed air, fluids or moving parts, never carry out maintenance and repair work with the engine running and:

- Make sure that nobody is standing in the path of moving parts before barring over or starting the engine.
- Use care when working around moving belts and rotating parts.
- Remove safety guards only to complete service procedures that can only be completed with the guard removed and replace safety guards as soon as possible once the service is completed. Do not crank or start the engine with the safety guards removed unless specifically required in the procedure.

The engine may need to be running for some types of diagnostic activities.

Let engine and exhaust cool down

All exhaust components are hot when the engine is running and remain hot for an extended period after the engine is shut off. Other components and fluids are hot when the engine is running and remain hot for an extended period after the engine is shut off. Allow all components and fluids to cool before working on or near the unit, or use proper precautions to reduce the risk of being burned.

Engine placement [QL3]

When placing an engine in an area to be serviced, remember the following to reduce your risk of injury or death:

- Place the engine on a firm, flat and stable surface capable of supporting the units weight. Unless authorized by an MTU dealer or distributor, never set an engine down on the oil pan.
- Before servicing anything raised by hydraulics, make sure that it is mechanically supported by stands, blocks or a mechanical lock. Never rely on a hydraulic circuit alone when working underneath raised equipment.

Safe Engine Servicing

Following safe work practices can help reduce the risk of serious injury or death to yourself or bystanders when servicing an engine.

Use proper emergency shutdown procedures

If you have an emergency shutdown situation, use the hot shutdown procedure. You may damage the engine. Each emergency situation is unique. Follow the restart procedures in the manual.

Use lockout and tagout procedures

To reduce the risk of severe injury or death from electric shock or unintentional startup, disconnect all electrical power sources and batteries. Additionally, lockout and tagout the equipment before removing protective shields for service or maintenance. Do not remove a lockout tag unless proper authorization is obtained.

Provide proper ventilation

Do not operate this engine in an area where combustible chemicals or vapors may be present. Combustible vapors near the air intake system could be ingested into the engine, causing the engine to suddenly accelerate and overspeed or explode. This condition could cause an unexpected increase in engine rpm or fire. When changing the engine oil or working on the fuel system, make sure that the service area is properly ventilated.

Avoid carbon monoxide poisoning

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